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Silicide-block-ring body layout for non-integrated body LDMOS and LDMOS-based lateral IGBT

Abstract

An integrated circuit includes a semiconductor substrate having a doped region, e.g. a DWELL, with a first conductivity type. A source region is located within the doped region, the source region having a second opposite conductivity type. A drain region having the second conductivity type is spaced apart from the source region. A gate electrode is located between the source region and the drain region, the gate electrode partially overlapping the doped region. A body region having the first conductivity type is located within the doped region. A dielectric layer forms a closed path around the body region.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application Ser. No. 63/043,199, filed on Jun. 24, 2020, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) This disclosure relates to the field of semiconductor devices, and more particularly, but not exclusively, to improved reliability and reduced size of transistor devices such as LDMOS and LDMOS-based IGBT.

BACKGROUND

(2) An insulated-gate bipolar transistor (IGBT) is a three-terminal power semiconductor device primarily used as an electronic switch. It combines high efficiency and fast switching. IGBTs switch electric power in many modern appliances: variable-frequency drives (VFDs), electric cars, trains, variable speed refrigerators, lamp ballasts, air-conditioners, and even stereo systems with switching amplifiers, etc. Since it is designed to turn on and off rapidly, amplifiers that use it often synthesize complex waveforms with pulse-width modulation and low-pass filters. In switching applications, modern devices feature pulse repetition rates well into the ultrasonic range-frequencies which are at least ten times the highest audio frequency handled by the device when used as an analog audio amplifier.

(3) The IGBT combines the simple gate-drive characteristics of a metal oxide semiconductor field effect transistor (MOSFET) with the high-current and low-saturation-voltage capability of a bipolar transistor. The IGBT combines an isolated-gate FET for the control input and a bipolar power transistor as a switch in a single device. Large IGBT modules typically include many devices in parallel and can have very high current-handling capabilities in the order of hundreds of amperes with blocking voltages of 6000 V. These IGBTs can control loads of hundreds of kilowatts.

(4) Various structures for IGBTs, such as: planar IGBTs, trench IGBTs, and lateral IGBTs, have been designed to customize the operational properties of the device for particular applications. For example, planar or vertical IGBTs utilize a convenient structure for a high power (e.g., high voltage and high current) switch. The planar IGBT includes a collector at a bottom side, a gate at a top side, and an emitter surrounding the gate at the top side. Trench gate IGBTs have a similar general structure to the planar IGBTs. However, trench IGBTs include a trench within which the gate is situated. The trench reduces the on-state voltage drop of the device. The current path of planar and trench IGBTs is vertical from the collector to the emitter.

(5) Lateral IGBTs (LIGBT) are often employed in lower power control and detection circuits. Lateral IGBTs do not utilize the vertical structure of the planar and trench IGBTs, where collector and emitter contacts are provided on the top and on the bottom of the semiconductor material. Instead, lateral IGBTs generally include a substrate contact at a bottom side, a collector at one side of a top side, an emitter at the other side of the top side, and a gate disposed between the emitter and the collector at the top side. The current path of lateral IGBTs is horizontal (e.g., lateral) within the device from the collector to the emitter.

SUMMARY

(6) Some examples provide semiconductor devices, and methods of making such devices. In one example an integrated circuit includes a semiconductor substrate having a doped region with a first conductivity type. A source region is located within the doped region, the source region having a second opposite conductivity type. A drain region having the second conductivity type is spaced apart from the source region. A gate electrode is located between the source region and the drain region, the gate electrode partially overlapping the doped region. A body region having the first conductivity type is located within the doped region. A dielectric layer forms a closed path around the body region.

(7) Other examples provide one or more methods of implementing the previously described examples.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a detailed description of various examples, reference will now be made to the accompanying drawings in which:
- (2) FIG. 1 illustrates an equivalent circuit schematic for a lateral IGBT, including a bipolar transistor and a MOS transistor;
- (3) FIG. 2 illustrates a lateral IGBT according to one example of the disclosure;
- (4) FIGS. 3A-3C illustrate a portion of a baseline IGBT that includes parallel SiBLK structures, each SiBLK structure located between a body region and a source region, including a resistance between the body region and a channel region;
- (5) FIGS. 4A-4D illustrate a portion of an example IGBT of the disclosure that includes parallel closed-loop SiBLK structures, each SiBLK structure circumscribing a body region, being separated from a neighboring SiBLK structure by a source region, and being located between two adjacent gate electrodes, including a smaller resistance between the body region and a channel region;
- (6) FIG. 5A-5D illustrate various examples of SiBLK structures that each form a closed path around a corresponding body region;
- (7) FIGS. 6A-6C illustrate section views through different locations of the example of FIG. 5A;
- (8) FIGS. 6D-6E illustrate section views through different locations of the example of FIG. 5B;
- (9) FIGS. 6F-6G illustrate section views through different locations of the example of FIG. 5C;
- (10) FIGS. 6H-6I illustrate section views through different locations of the example of FIG. 5D;
- (11) FIG. 7 illustrates normalized unit current per length for devices fabricated according to the baseline configuration of FIG. 3B, and the example configurations of FIGS. 5A-5D; and
- (12) FIG. 8 illustrates normalized $I_{\text{sub.DSAT}}$ per unit area for devices fabricated according to the baseline configuration of FIG. 3B, and the example configurations of FIGS. 5A-5D; and
- (13) FIG. 9A-9J each show a section view through a circumscribed body region, and a source region, of an example lateral IGBT of the disclosure at various stages of manufacturing.

DETAILED DESCRIPTION

- (14) The present disclosure is described with reference to the attached figures. The figures are not necessarily drawn to scale, and they are provided without implied limitation to illustrate various described examples. Several aspects of the disclosure are described below with reference to example applications for illustration. It should be understood that numerous specific details, relationships, and methods are set forth to provide an understanding of the disclosure. The present disclosure is not limited by the illustrated ordering of acts or events, as some acts may occur in different orders and/or concurrently with other acts or events. Furthermore, not all illustrated acts or events are required to implement a methodology in accordance with the present disclosure.
- (15) Toward a goal of increasing current efficiency and safe operating area (SOA) in some transistor devices, some examples of the present disclosure reduce the body resistance of such transistors by reducing the distance between the body and the source. The reduction is made possible by providing a new configuration of a dielectric layer used to conductively isolate the body from the source at a surface of the substrate in which the transistor is implemented. In such a configuration the dielectric layer forms closed loops, each closed loop enclosing a corresponding body region. Source regions, and metal contacts to the source regions, are located between nearest-neighbor body regions. While such configurations may increase current efficiency and SOA, no particular result is a requirement of the present invention unless explicitly recited in a particular claim.
- (16) While the described examples include specific combinations of N-type regions and P-type regions, those skilled in the art will appreciate that the conductivity types of the various doped regions may be reversed to form analogous examples that fall within the scope of the disclosure.
- (17) FIG. 1 illustrates an equivalent circuit schematic of an integrated circuit (IC), including an electronic device **100**. Without implied limitation, the IC may be designed to control power

between a source and a load, or similar applications. The electronic device **100** is sometimes referred to herein without implied limitation as an insulated-gate bipolar transistor (IGBT). The device **100** includes a bipolar junction transistor (BJT) **101** and an extended drain (ED) MOSFET **102**. A drain terminal **105** of the MOSFET **102** is connected to a base **110** of the BJT **101**. An extended drain terminal **115** of the MOSFET **102** and an emitter **120** of the BJT **101** are connected at an anode (A) **125** of the device **100**. In a nonlimiting configuration, a diode **103** is connected between a collector **130** of the BJT **101** and a source **135** of the MOSFET **102**, with the node between the collector **130** and the diode **103** serving as a cathode (K) **137**. A resistor **108** represents a resistance R_D associated with an extended drain of the MOSFET **102**, and a resistor R_B represents a resistance associated with a body resistance between a body **160** of the MOSFET **102** and the collector **130**. A gate **145** provides a control terminal to place the device **100** in a conduction state between the anode **125** and the cathode **137**, or in a high-impedance state.

(18) FIG. 2 presents a plan view of an electronic device **200** that may implement the device **100** according to various examples of the disclosure. The device **200** may be referred to without implied limitation as a transistor **200**, while recognizing that more than one transistor may operate within the device **200**. The example transistor **200** may be described as a multi-finger transistor, in which N gate electrodes **245**, three in the present example, may be considered as fingers of the device. The source **135** is implemented by $N+1$ source regions **250**, sometimes collectively referred to as the source **250**. An instance of the gate electrode **245**, sometimes referred to herein simply as a gate **245**, is located between nearest-neighbor instances of the source regions **250**. A drain region **215** corresponding to the extended drain terminal **115** is accessed through two openings in each of LOCOS layers **222**, and an emitter region **225** is accessed through a single opening in each LOCOS layer **222** between the drain region **215** openings. In some other examples, shallow-trench isolation (STI) structures may be used in lieu of the LOCOS layers **222**. The drain region **215** is physically implemented with an $N_{\text{sup.}}$ region that may be referred to equivalently as a drain **215**. The gate **245** partially overlaps nearest-neighbor instances of the source region **250**. Optional dummy gates **255** spaced apart from first and last instances of the gates **245** may be included to improve manufacturing uniformity of features within the area circumscribed by the source regions **250**. Various metal vias, generally referred to as contacts **240**, provide conductive connections to the various structures of the transistor **200**.

(19) Between the gates **245** are located a number of body regions **260**, collectively referred to as the body region **260**. A corresponding dielectric structure **265** encloses, surrounds, or circumscribes each body region **260**. A silicide layer, not explicitly shown, overlies the source regions **250** and the body regions **260**, e.g. to provide an ohmic connection between a metal contact and the underlying semiconductor. The dielectric structures **265** prevent formation of the silicide layer at their locations, thus ensuring the body regions **260** are not directly conductively connected to the source regions **250**. This property of the dielectric structures **265** is sometimes referred to in the art as “silicide blocking”, sometimes shortened to “SiBLK”. As such the dielectric structures **265** may be equivalently referred to herein as SiBLK **265**.

(20) FIG. 3A shows a section view of a baseline transistor **300** that is similar to the transistor **200**, but differs in certain respects as described below. A substrate **305** supports a layer **317**, such as a lightly-doped epitaxial layer, having a first conductivity type, or P-type. A drift region **330** is located between two P-type double-diffused wells, or DWELLS, **335**. Body regions **360**, each located between two source regions **350**, are located in each DWELL **335**. A drain region **315** having a second conductivity type, or N-type, may act as an anode when connected to a $P_{\text{sup.}}$ emitter contact (not shown), and is located within the drift region **330**. A LOCOS **322** structure surrounds the drain region **315**. A gate electrode **345** extends from the drain region **315** to a source region **350** in each of the DWELLS **335**, and surrounds the drain region **315**. Unreferenced metal contacts provide vertical connections to the source regions **350**, body regions **360** and drain region **315**. In particular, a contact **340** to the source region **350** is discussed in greater detail below. An

SiBLK layer **365**, sometimes referred to as SiBLK **365**, is located over each junction formed between one of the body regions **360** and one of the source regions **350**.

(21) FIG. **3B** illustrates a plan view of a portion of the transistor **300** taken between the gate electrode **345** and an adjacent gate electrode **345** as marked in FIG. **3A**. The illustrated view includes one body region **360** between two source regions **350**. An optional source extension **353** extends from each of the source regions **350** under unreferenced dielectric spacers on sidewalls of the gate electrodes. The body region **360** and the two source regions **350** are arranged as parallel stripes. SiBLK **365** lines, also arranged as parallel stripes, overlie the junctions between the body region **360** and the source regions **350**. This configuration is representative of some baseline IGBT transistors that have a greater body resistance than desired.

(22) Focusing on the vicinity of the source region **350**, FIG. **3C** illustrates a body resistance $R_{sub.B1}$ between the body region **360** and a channel region **355** under the gate electrode **345**. The value of $R_{sub.B1}$ is in part a function of the width of the source region **350** that spaces the body region **360** apart from the channel region **355**. The width of the source region **350** in turn depends in part on the presence of the contact **340**. The source region **350** width must accommodate the width of the contact **340**. But also, because technology design rules enforce minimum spacing between the contact **340** and the SiBLK **365**, and between the contact **340** and the gate electrode **345**, the minimum width of the source region **350** must also accommodate such requirements. Thus the value of $R_{sub.B1}$ cannot be reduced below a lower limit determined by such factors.

(23) The value of the body resistance generally impacts device performance by allowing a parasitic NPN transistor formed by the source region **350**, the DWELL **335** and the drain region **315** to trigger at a lower voltage, thus reducing the safe operating area (SOA) of the transistor. Furthermore, the width of the source region **350** needed to accommodate the contact **340** results in a larger half-pitch of the transistor, represented by $HP_{sub.1}$ in FIG. **2A**. The larger half-pitch increases die area of the substrate on which the transistor is formed.

(24) Returning to FIG. **2**, the transistor **200** addresses some deficiencies of the baseline transistor **300** by segmenting the body into the body regions **260**, and placing contacts to the source regions **250** between the body regions **260**, allowing the body regions to be located closer to the channel regions of the transistor **200**. FIGS. **4A-4C** present three cut locations as marked in FIG. **2** to further illustrate this aspect of the transistor **200**.

(25) FIG. **4A** shows a section view through the emitter region **225** of the transistor **200** and neighboring instances of the body region **260**. A P-type layer **217**, e.g. a lightly-doped epitaxial layer, overlies a substrate **205**. The emitter region **225** is located within an N-drift region **208** that in turn is located within the layer **217**. The source regions **250** and body regions **260** are located within DWELLS **235**. Source extensions **253** extend from each source region **250** under unreferenced dielectric spacers on the sidewalls of the gate electrodes **245**. The source extensions **253** are described in greater detail below. A detail view in FIG. **4D** shows salient features of the source region **250**, one instance of the body region **260**, and one instance of a gate **245**. A body resistance value $R_{sub.B2}$ characterizes the resistance between the body region **260** and a channel region **256** under the gate **245**. The value of $R_{sub.B2}$ is determined in part by the width of the source region **250**, which now neither includes the width of the contact **340** (FIG. **3C**), nor the minimum spacing rules associated with the contact **340**. Because the width of the source region **250** (FIG. **4D**) is lower than the width of the source region **350** (FIG. **3C**), the value of $R_{sub.B2}$ is less than the value of $R_{sub.B1}$, resulting in improved SOA. Moreover, because the width of the source region **250** is less than the width of the source region **350**, a half-pitch of the transistor **200**, represented by $HP_{sub.2}$, is less than the half-pitch $HP_{sub.1}$ of the transistor **300** (FIG. **3A**), resulting in smaller die area needed to implement the transistor **200** relative to the baseline transistor. (The half-pitch of the transistor **200** may be determined by N multiples of HP_1 (FIG. **3A**) plus additional width related to the dummy gates **255**, if present.)

(26) FIG. **4B** illustrates another sectional view of the transistor **200**, taken through the gate **245** and

two adjacent source regions **250**. The distance between neighboring gates **245** may be determined at least in part by the minimum dimensions and spacing of the body region **260** (including contacts), and the width of the SiBLK **265** (FIG. **4A**). The number of contacts to the source region **250** between the gates **245**, shown as three in the example of FIG. **4B**, may be determined by design rule constraints regarding contact width and spacing in the available gap between the adjacent gates **245**. For completeness, FIG. **4C** illustrates another sectional view of the transistor **200**, taken through the drain region **215** and an instance of the body region **260**. Unreferenced metal contacts to the body regions **260** are circumscribed by the SiBLK **265**, as seen more clearly in FIG. **1A**.

(27) FIGS. **5A-5D** illustrate various configurations of the body region **260**, the SiBLK **265**, and adjacent gates **245**. FIGS. **6A-6I** provide illustrative cross-sections of indicated locations in FIGS. **5A-5D**. FIGS. **6A-6I** are presented with a larger horizontal scaling than that of FIGS. **5A-5D** to better show some structural details of these examples.

(28) FIG. **5A** shows for reference the configuration previously illustrated in FIG. **2**, now designated configuration **510**. The body regions **260** in the illustrated example have a rectangular shape. More generally the body regions have sides that are parallel to the edges of the gates **245**, with those parallel sides being joined by ends that may have any outline, e.g. straight or rounded. In some examples the body regions have a quadrilateral shape with 90° corners, such as a square or rectangle, but could in principle be trapezoidal, or be parallelograms with corners being other than 90°. The remaining discussion assumes without implied limitation that the body regions **260** are rectangular.

(29) FIG. **6A** shows a section view taken along the long axis of the source region **250** and through the body regions **260**, while FIG. **6B** shows a section view perpendicular to the view of FIG. **6A**, through the body region **260**. FIG. **6B** is analogous to FIG. **4A**, but centered on the body region **260**. FIG. **6A** illustrates the alternate P+ doping of the body regions **260** touching the N+ doping of the source region **250**. Sections of the SiBLK **265**, which surround the body regions **260**, are shown located over the junction formed between the body regions **260** and the source region **250**. FIG. **6B** shows a section view viewed at the body region **260**, and FIG. **6C** shows a section view viewed at the source region **250**. In both of these views the DWELL **235** and the source region **250** both extend under the gates **245**. The body region **260** and the SiBLK **265** are present in FIG. **6B** and absent in FIG. **6C**.

(30) FIG. **5B** shows an example configuration **520** in which a source region **250'** has a width equal to the width of the body region **260**, with source extensions **253** extending between the source regions **250'** and the gates **245**. FIG. **6D** illustrates a section view through the body region **260**, while FIG. **6E** illustrates a section view through the source region **250'**. As compared to the configuration **510**, the configuration **520** spaces the source region **250'** apart from the gate **245** such that the P-type layer **217** intersects the surface between the gate **245** and the source region **250'**.

(31) FIG. **5C** shows an example configuration **530** in which the distance between the gates **245** is reduced relative to the configuration **510**. A source region **250''** is wider than the body region **260**, and extends under the gates **245**. FIG. **6F** illustrates a section view through the body region **260**, while FIG. **6G** illustrates a section view through the source region **250''**. The body region **260** has the same width relative to the configuration **510**, resulting in the spacing between the body region **260** and the gates to be reduced. Further, the SiBLK **265** overlaps the gates **245**. Thus the configuration **530** has a half-pitch smaller than that of the configurations **510**, **520**.

(32) Finally FIG. **5D** shows an example configuration **540** that is similar to the configuration **530** in that the distance between the gates **245** is reduced relative to the configurations **510** and **520**. The configuration **540** is also similar to the configuration **520** in that a source region **250'''** has the same width as the body region **260**, with source extensions **253** extending between the source regions **250'''** and the gates **245**. FIG. **6H** illustrates a section view through the body region **260**, while FIG. **6I** illustrates a section view through the source region **250'''**. Thus the distance between the body

region **260** and the gate **245** is again reduced relative to the configuration **510**. And similar to the configuration **520**, portions of the layer **217** intersect the substrate surface between the source region **250'''** and the gates **245**. The configuration **540** has the same half-pitch as does the configuration **530**.

(33) Table I displays relevant dimensional attributes of a representative baseline configuration characterized by FIGS. **3A** and **3B**, and example configurations of the disclosure characterized by FIGS. **5A** and **5B**, and FIGS. **5C** and **5D**. In these examples, presented without implied limitation, the spacing from the body region **260** to the gates **245** is about $0.375\text{ }\mu\text{m}$ for the examples of FIGS. **5A** and **5B**. Conversely, the spacing from the body region **260** to the gates **245** is about $0.150\text{ }\mu\text{m}$ for the examples of FIGS. **5C** and **5D**. These spacing values reflect the minimum spacing allowed by the applicable design rules of the process technology used to design these transistors. Thus, whereas minimum design rules result in a gate-to-body spacing of $0.675\text{ }\mu\text{m}$ for the baseline transistor **300**, examples consistent with the disclosure allow a gate-to-body spacing less than $0.5\text{ }\mu\text{m}$. These values demonstrate that the spacing allowed by the example layouts of FIGS. **5A-5D** may result in reduction of transistor half-pitch by 10% to 20%.

(34) TABLE-US-00001 TABLE I Layout style Gate-to-body spacing Half pitch Baseline $0.675\text{ }\mu\text{m}$ $2.85\text{ }\mu\text{m}$ FIGS. **5A/5B** $0.375\text{ }\mu\text{m}$ $2.55\text{ }\mu\text{m}$ FIGS. **5C/5D** $0.15\text{ }\mu\text{m}$ $2.25\text{ }\mu\text{m}$

(35) Example IGBT transistors with the general form of FIG. **2** were manufactured using each of the configurations **510-540**, and tested electrically to determine the effect of these configurations on SOA. FIG. **7** illustrates transmission line pulse (TLP) current versus TLP voltage, and FIG. **8** presents $I_{\text{sub.DSAT}}$ (saturation drain current) normalized by transistor area versus snapback voltage $V_{\text{sub.SNAP}}$. These data demonstrate that the configuration **520** (FIGS. **5B**, **6D** and **6E**) results in an improvement of current efficiency of at least a 20% relative to the baseline configuration **300** (FIGS. **3A** and **3B**). The configuration **540** (FIGS. **5D**, **6H** and **6I**) provides an intermediate improvement of current efficiency with respect to the baseline configuration.

(36) The reduction of transistor half-pitch and the improvement of current efficiency provided by examples described herein, especially of the configuration **520**, demonstrate a significant and valuable improvement of IGBT device layout. The principles described herein do not require a minimum design rule layout, and the gate-to-body spacing is not limited to any particular value. But as a practical matter a spacing of $1\text{ }\mu\text{m}$ or less is expected to provide the observed improved characteristics of the transistor **200** while providing a compact design.

(37) FIGS. **9A-9J** illustrate an example method **900** that may be used as part of the formation of the transistor **200**. The method **900** is but one of several possible sequences of process steps, and those skilled in the art will appreciate that some steps may be reordered, e.g. to accommodate a particular process technology or tool set. The configuration **520** is used as a specific example, but those skilled in the pertinent art will appreciate that the method **900** can be adapted to form the configurations **510**, **530** and **540**, and similar variations within the scope of the disclosure, without undue experimentation. In each of these figures, a section view is provided through the transistor **200** at the location at which the body region **260** is formed, and the location at which the source region **250** is formed. The method **900** is described without implied limitation in terms of some regions doped P-type and others doped N-type. Those skilled in the art will recognize that the method may be adapted to reverse the dopant types of the various regions to realize an analogous device.

(38) In FIG. **9A** the substrate **205** is provided having the epitaxial layer **217** located thereover. The substrate **205** may optionally include a buried layer, such as an N-buried layer, or a buried insulating layer, such as an SOI layer. Optionally the epitaxial layer **217** is lightly doped P-type.

(39) In FIG. **9B**, N-drift regions **208** have been formed, e.g. by a moat implant. The formation may include conventional steps such as photoresist deposition and patterning, and ion implantation of an N-type dopant such as phosphorous to produce a dopant concentration in a range between about $1\text{e}16\text{ cm}^{-3}$ and about $1\text{e}18\text{ cm}^{-3}$. The dopant may have been diffused in an anneal step

prior to this figure, or may be done in a later step at which multiple implanted regions are annealed concurrently.

(40) In FIG. 9C, the DWELL 235 has been implanted with a P-type dopant such as boron, and diffused into the layer 217. The DWELL 235 illustrated in this figure is representative of a DWELL that may be formed by one or more implants that may be performed before or after gate poly formation (e.g. FIG. 9E). The dopant concentration in the DWELL 235 by the one or more implants may be in a range between about $1 \times 10^{17} \text{ cm}^{-3}$ and about $1 \times 10^{19} \text{ cm}^{-3}$.

(41) FIG. 9D shows the partially formed transistor 200 after forming LOCOS layers 222. These structures may be formed by a conventional sequence including forming and patterning a silicon nitride layer over the epitaxial layer 217, followed by steam oxidation of exposed portions of the exposed surface. The silicon nitride layer is then removed, after which a thermal oxidation may be performed to form a gate dielectric layer (not shown) at the top surface of the layer 217. In alternative examples, not shown, STI structures may be used in lieu of the LOCOS layers 222.

(42) FIG. 9E shows the partially formed transistor 200 after a polysilicon layer has been formed over the doped region 235 and patterned to form the gate electrodes 245, which may be done conventionally.

(43) FIG. 9F shows the partially formed transistor 200 after an N-type dopant such as phosphorous has been implanted in the DWELL 235. The implant may be self-aligned, e.g. using the gate electrode 245 as a masking pattern. The partially formed transistor 200 has been annealed to diffuse the N-type dopant into the DWELL 235, which may further diffuse the P-type dopant of the DWELL 235 into the epitaxial layer 217. The diffusing results in the DWELL 235 having the dopant concentration previously described, and the source extension 253 having a dopant concentration in a range between about $1 \times 10^{19} \text{ cm}^{-3}$ and about $1 \times 10^{20} \text{ cm}^{-3}$. This doping level is represented symbolically by “N.sup.+” for convenience. Because the source extension 253 is self-aligned to the end of the gates 245, the source extension 253 may extend partially under the gates 245 after the diffusion.

(44) FIG. 9G shows the partially formed transistor 200 after forming unreference dielectric spacers on sidewalls of the gate electrodes 245. The sidewall spacers may be formed conventionally, and may include one or more dielectric layers, e.g. silicon oxide and/or silicon nitride. Further, a masked implant of a P-type dopant such as boron into the DWELL 235 has been performed, thereby forming the body region 260, and masked implant of an N-type dopant such as phosphorous into the DWELL 235 has been performed, thereby forming the source region 250'. These dopants may be implanted using process conditions similar to implanting P-source/drain (PSD) and N-source-drain (NSD) in a CMOS process technology, and may be referred to for convenience as PSD and NSD implants. The body region 260 and the source region 250' may each have a concentration of their respective dopants in a range between about $1 \times 10^{20} \text{ cm}^{-3}$ and about $1 \times 10^{21} \text{ cm}^{-3}$. These doping levels are respectively represented symbolically by “N.sup.++” and “P.sup.++” for convenience.

(45) In FIG. 9H a dielectric layer 910 is formed over the gate electrodes 245, epitaxial layer 217 and sidewall spacers. The layer 910, from which the SiBLK 265 features are formed, may be any suitable dielectric layer or layers, typically characterized by being capable of being selectively removed without significant removal of other material features, such as the sidewall spacers. In some examples the layer 910 includes nitrogen, such as silicon nitride or silicon oxynitride. In some other examples the layer 910 is substantially free of nitrogen, e.g. silicon oxide. Regardless of the material type, the layer 910 is not limited to a thickness of any particular value, but may have a thickness in a range between about 20 nm and 100 nm, inclusive. A photoresist pattern (not shown) is formed over the layer 910, e.g. by conventional photolithography, to protect locations at which the dielectric structures 265 are to be formed.

(46) FIG. 9I shows the partially formed transistor 200 after selective removal of the layer 910, and cleanup, resulting in the SiBLK 265 features as previously described. Finally, FIG. 9J illustrates the

partially formed transistor **200** after forming a silicide layer **920** on exposed portions of the gate electrodes **245**, source region **250'** and the body region **260**. Additional processing steps, not shown, are performed to provide a dielectric layer over the gate electrodes **245**, SiBLK **265**, body region **260** and source region **250'**. Metal contacts and interconnections are formed in or over the dielectric layer to complete the transistor **200**.

(47) The above discussion is meant to be illustrative of the principles and various embodiments of the present invention. Numerous variations and modifications will become apparent to those skilled in the art once the above disclosure is fully appreciated. It is intended that the following claims be interpreted to embrace all such variations and modifications.

Claims

1. An integrated circuit, comprising: a semiconductor substrate having a doped region with a first conductivity type; a source region located within the doped region, the source region having an opposite second conductivity type; a drain region having the second conductivity type spaced apart from the source region; and a gate electrode between the source region and the drain region, the gate electrode partially overlapping the doped region; a heavily-doped body region having the first conductivity type located within the doped region and directly touching the source region; and a first dielectric layer that forms a closed path around the body region and extends along a top surface of the source region and a top surface of the body region over a junction between the source region and the body region; and a second dielectric layer that laterally abuts the first dielectric layer and is located on the top surface of the source region and the top surface of the body region.
2. The integrated circuit of claim 1, wherein the first dielectric layer comprises silicon oxynitride.
3. The integrated circuit of claim 1, wherein the first dielectric layer encloses a rectangular area.
4. The integrated circuit of claim 1, wherein a source extension is located between the source region and the gate electrode, the source extension having a lower dopant concentration than the source region.
5. The integrated circuit of claim 1, wherein the closed path is a first closed path enclosing a first body region, and the first dielectric layer forms a second closed path enclosing a second body region, further comprising a plurality of metal contacts conductively connected to the source region directly between the first and second closed paths.
6. The integrated circuit of claim 1, wherein a space between the gate electrode and the body region is less than 1 μm .
7. The integrated circuit of claim 1, wherein the source region, the drain region and the gate electrode are components of an insulated-gate bipolar transistor.
8. The integrated circuit of claim 1, wherein the first conductivity type is P-type and the second conductivity type is N-type.
9. A method of manufacturing an integrated circuit, comprising: forming a doped region having a first conductivity type in a semiconductor substrate; forming a source region within the doped region, the source region having an opposite second conductivity type; forming a drain region having the second conductivity type spaced apart from the source region; forming a gate electrode between the source region and the drain region, the gate electrode partially overlapping the doped region; forming a body region having the first conductivity type within the doped region; forming a first dielectric layer that forms a closed path around the body region and is located on a top surface of the source region and a top surface of the body region and touches a junction between the source region and the body region; and forming a second dielectric layer that laterally abuts the first dielectric layer and is located on the top surface of the source region and the top surface of the body region.
10. The method of claim 9, wherein the first dielectric layer comprises silicon nitride.
11. The method of claim 9, wherein the first dielectric layer encloses a rectangular area.

12. The method of claim 9, wherein the body region touches the source region.
 13. The method of claim 9, wherein the closed path is a first closed path enclosing a first body region, and the first dielectric layer forms a second closed path enclosing a second body region, further comprising forming a plurality of metal contacts conductively connected to the source region directly between the first and second closed paths.
 14. The method of claim 9, wherein a space between the gate electrode and the body region is less than 0.5 μm .
 15. The method of claim 9, wherein the source region, the drain region and the gate electrode are components of an insulated-gate bipolar transistor.
 16. The method of claim 9, wherein the first conductivity type is p-type and the second conductivity type is n-type.
 17. An integrated circuit, comprising: an epitaxial layer located over a semiconductor substrate and having a first conductivity type; first and second drift regions within the epitaxial layer and having an opposite second conductivity type; a doped region having the first conductivity type within the epitaxial layer between the first and second drift regions; first and second drain regions having the second conductivity type respectively located in the first and second drift regions; and first and second source regions having the second conductivity type and spaced apart within the doped region; a first gate electrode between the first drain region and the first source region, and a second gate electrode between the second drain region and the second source region; a heavily-doped body region having the first conductivity type located within the doped region and between the first and second source regions, and forming a first junction with the first source region and a second junction with the second source region; a silicide blocking layer that forms a closed path around the body region and is located on top surfaces of the first and second source regions and a top surface of the body region and overlaps the first and second junctions; and a dielectric layer over the silicide blocking layer that is located on the top surfaces of the first and second source regions and the top surface of the body region, wherein the first and second source regions provide a source contact of an insulated-gate bipolar transistor (IGBT), and the body region provides a collector contact of the IGBT.
 18. The integrated circuit of claim 17, wherein the silicide blocking layer comprises silicon nitride.
 19. An integrated circuit, comprising: a semiconductor substrate having a doped region with a first conductivity type; a source region located within the doped region, the source region having an opposite second conductivity type; a drain region having the second conductivity type spaced apart from the source region; and a gate electrode between the source region and the drain region, the gate electrode partially overlapping the doped region; a heavily-doped body region having the first conductivity type located within the doped region and directly touching the source region; and a dielectric layer that forms a closed path around the body region and a contiguous portion of the dielectric layer overlies the gate electrode and is directly on the body region.
 20. The integrated circuit of claim 19, wherein the source region provides a source contact of an insulated-gate bipolar transistor (IGBT), and the body region provides a collector contact of the IGBT.
 21. An integrated circuit, comprising: a semiconductor substrate having a doped region with a first conductivity type; a source region located within the doped region, the source region having an opposite second conductivity type; a drain region having the second conductivity type spaced apart from the source region; and a gate electrode between the source region and the drain region, the gate electrode partially overlapping the doped region; a heavily-doped body region having the first conductivity type located within the doped region and directly touching the source region; and a contiguous dielectric layer that forms a closed path around the body region and extends along a top surface of the body region and over the gate electrode.
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